ABES ENGINEERING COLLEGE, GHAZIABAD

Department of Electronics & Communication Engineering

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Mini Project Technical Report (B.Tech 3rd Year)

Project: Autonomous IoT Decision and Alert Management Platform (AIDAMS)

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Repository	github.com/prakashchinmay2211-bot/iot-autonomous-alert-platform

1. Abstract

The Autonomous IoT Decision and Alert Platform (AIDAMS) is an intelligent, edge-to-cloud industrial monitoring system designed to eliminate human reaction latency and alarm fatigue in mission-critical operations. Factory assets (bottle filling stations, CNC milling centers, logistics conveyors, and hydraulic pump turbines) often suffer catastrophic breakdown due to rapid thermal runaway or vibration surges. AIDAMS streams multi-sensor telemetry over lightweight MQTT (QoS-1), processes it via an unsupervised Isolation Forest anomaly detector and rolling Z-score statistical engine, and automates protective closed-loop control actions (emergency breaker trips, cooling boosts, speed throttling) in under 2 seconds. All telemetry, evaluations, decisions, and alerts are persisted in MongoDB. A live web cockpit with Plotly charts, an audit trail, and an in-app Rule Editor lets operators govern thresholds in real-time.

2. Problem Statement

Modern smart factories generate massive streams of telemetry, yet supervision remains predominantly reactive. High-speed rotating machinery experiences structural destruction in 2 to 5 seconds during lubrication or thermal failure—far faster than a human can respond. Furthermore, static threshold alarms cause severe false alarm fatigue during ambient shifts, and proprietary SCADA systems prohibit custom ML deployment while costing thousands in recurring cloud fees.

3. Project Objectives

• Ingest 6 telemetry streams (Temp, Vib, Pressure, Current, Noise, RPM) over MQTT.
• Deploy Isolation Forest for unsupervised zero-day anomaly detection without labeled datasets.
• Predict 5 failure modes and compute Equipment Health Index (0–100%).
• Execute closed-loop protective actions (Breaker Cutoff, Cooling Boost, Speed Throttling) in under 2 seconds.
• Provide a user-configurable rule management cockpit where operators can add/tune rules live.
• Integrate Node-RED workflows and Grafana Cloud dashboards for 24/7 free global accessibility.

4. Functional Requirements Specification

Req ID	Module	Requirement Specification
FR-01	MQTT Ingestion	Stream continuous multi-sensor telemetry over topic iot/telemetry/#
FR-02	Normalization	Standardize features via Z-score scaling and persist to MongoDB telemetry
FR-03	Anomaly Detection	Apply Isolation Forest to compute continuous anomaly score and flags
FR-04	Health Scoring	Compute Equipment Health Index (0-100%) and multi-class failure probabilities
FR-05	Autonomous Actuation	Trigger emergency breaker trip, cooling boost, and speed throttling
FR-06	Debounce Logic	Enforce 30s cooldown per rule to prevent electromechanical relay chatter
FR-07	Rule Governance	Allow operators to add, edit, toggle, and delete decision rules live in the UI
FR-08	Explainable AI	Identify top driving sensor deviation (%) and plain-language root causes
FR-09	Incident Lifecycle	Track alert states (ACTIVE -> ACKNOWLEDGED -> RESOLVED) with logs
FR-10	Webhooks & Alerts	Dispatch automated JSON payloads to Discord, Slack, and Telegram Bot

5. Autonomous Protective Actions Matrix

Target Asset	Trigger Condition	Autonomous Action	Physical Countermeasure
MCH-007 (Bottle Filling)	Temp >= 88C OR Vib >= 4.0 mm/s	EMERGENCY_FAULT_STOP	Fill valve shut, conveyor to 0, cooling to 100%
CNC-003 (Milling)	Spindle Temp > 85C OR Vib > 5.5	SPINDLE_SAFETY_TRIP	Spindle brake on, feed rate 0, flood coolant 100%
CONV-012 (Conveyor)	Motor Temp > 80C OR Tension > 620N	EMERGENCY_CONVEYOR_HALT	Drive motor halted, diverter opened to bypass
ENV-001 (Air Quality)	Gas > 250 ppm OR Smoke > 80 ppm	HAZMAT_EVACUATION_PURGE	Air intake shut, exhaust fans 100%, siren active
Generic Turbine	Health < 45% OR Temp > 90C with ML	EMERGENCY_SHUTDOWN	Main breaker tripped offline, cooling boosted

6. System Verification & Test Execution Results

• Test TC-01 (Module Imports): All ML, Decision Engine, and MQTT components initialized cleanly.
• Test TC-02 (ML Diagnostic Fitting): Random Forest trained with 100.0% accuracy on 2,500 samples.
• Test TC-03 (Baseline Operation): Nominal readings evaluated at Health Index = 79.8%.
• Test TC-04 (Thermal Surge Fault): Overheating injected; health dropped to 12.0%, Emergency Shutdown engaged.
• Test TC-05 (Debounce Anti-Flapping): Successive anomalous packet blocked during 30s cooldown.
• Test TC-06 (Node-RED & Grafana): Flows and dashboard schemas validated for 1-click cloud import.

7. References

1. HCL Career Shaper AIoT Curriculum (Lectures 4–7) & Architecture Guide for Industrial Automation.
2. Liu, F. T., Ting, K. M., & Zhou, Z.-H. (2008). 'Isolation Forest.' IEEE ICDM, pp. 413-422.
3. Pedregosa et al. (2011). 'Scikit-learn: Machine Learning in Python.' JMLR 12, pp. 2825-2830.
4. OASIS Standard (2019). 'MQTT Version 5.0.' OASIS Open Standards.
5. Streamlit Community Cloud Documentation (2026). https://docs.streamlit.io